MobileAgents: Designing and Evaluating a Mobile Multimodal Interface for Controlling Personal AI Agents On the Go

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Abstract

People increasingly rely on personal AI agents—for research, messaging, scheduling—but controlling these agents away from a desktop is cumbersome: typing complex instructions on a phone is slow, and existing orchestration frameworks offer no end-user interface at all. We designed MobileAgents, a mobile-first platform where users bring their own AI agents and control them on the go through text, voice, and camera input. An LLM orchestrator decomposes multimodal requests into a dependency-aware plan across the user's personal agent pool, and a live DAG streams execution progress. The architecture is agent-agnostic: any agent can be registered with its role and tools at runtime—no code changes required. To study how much of this orchestration to reveal, we compared three transparency configurations—Black Box, Plan Preview, and Full Transparency—in a mixed-methods study with 13 participants (think-aloud, surveys, interviews). Plan Preview offered the best balance of trust and efficiency for mobile use; voice and image input were valued for on-the-go scenarios but required consent gating. We discuss value trade-offs, automation bias risks, and design implications for mobile multimodal agent control.

CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**; **Empirical studies in HCI**.

Keywords

multimodal interaction, agent orchestration, transparency, human-in-the-loop, trust, human–AI interaction

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1 Introduction: Use Case and Goals

AI Agents are increasingly capable of performing personal and professional tasks [12], which is motivating people to assemble personalized teams of agents. However, currently these agents need to be controlled via a programming interface or specialized desktop applications. However, the main value of these agents is that they can serve us semi- or fully autonomously while we are away from the workplace or a desktop computer. This has created strong demand for mobile applications that allow people to control their own agents while away from their workstations. This was demonstrated by the strong response to ClawdBot [**?**], an open-source autonomous AI agent controlled via messaging platforms such as WhatsApp. Yet Clawdbot operates as a single agent with difficult-to-control permissions, planning, and approval. Additionally, existing agentic management platforms, such as OpenAI Frontier and blah blah, are both commercial and not optimized for mobile use while on the go. This underscores the need for a fully open-source, fully mobile agent management platform that enables people to manage their custom agentic workflows with adjustable levels of control and governance while on the go.

We designed a mobile-first platform that lets users *bring their own AI agents* and control them on the go through text, voice, and images (Figure 2). The architecture is **agent-agnostic**. An LLM orchestrator breaks down requests, assigns tasks to agents, and waits for them to complete before returning a cohesive result. Multimodality is key in this design. Text is sent via a standard conversational interface that supports asynchronous communication with agents. Voice enables hands-free control while walking or cycling. Images allow you to snap a photo in the environment and send it as context to the agents. Given that these workflows may execute tasks with varying impact and risks, we prioritize having a highly customizable level of control. Certain tasks will be executed completely autonomously, while others may require approval, additional input, or real-time monitoring. [INSERT HOW THIS APP Is evaluated]. Multimodality and control are essential not only for providing the necessary flexibility to assemble custom multi-agent systems but also to enable mobility and easy-to-use interaction.

We envision that mobile control of custom multi-agent systems will be essential in modern human-AI interaction and key to effective management of personal and professional agents. This open-source application is initially targeted at people with a technical background who already use agents in their day-to-day life. Examples may include academics, managing agents for explorative research, summarization, and documentation, software engineers remotely steering teams of long-running coding agents, or quantitative analysts who use agents for trading and financial data analysis. As agents become increasingly capable in more domains and familiar to non-technical users, the design of this application can be expanded to target anyone who utilizes AI.

2 Methods: System Design and Implementation

2.1 Prototype Overview

MobileAgents is a fully functional prototype (Figure 2; full architecture in Appendix A–B).

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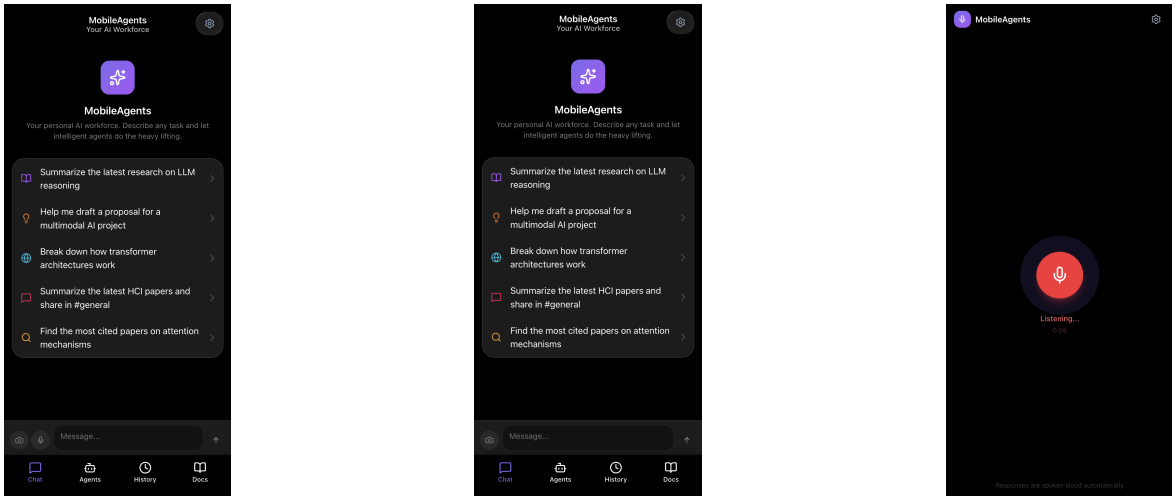


(a) Multimodal mode (b) Text + Images mode (c) Voice-only mode (listening)

Figure 1: MobileAgents input modes. (a) Multimodal (b) Text + Images. (c) Voice-only

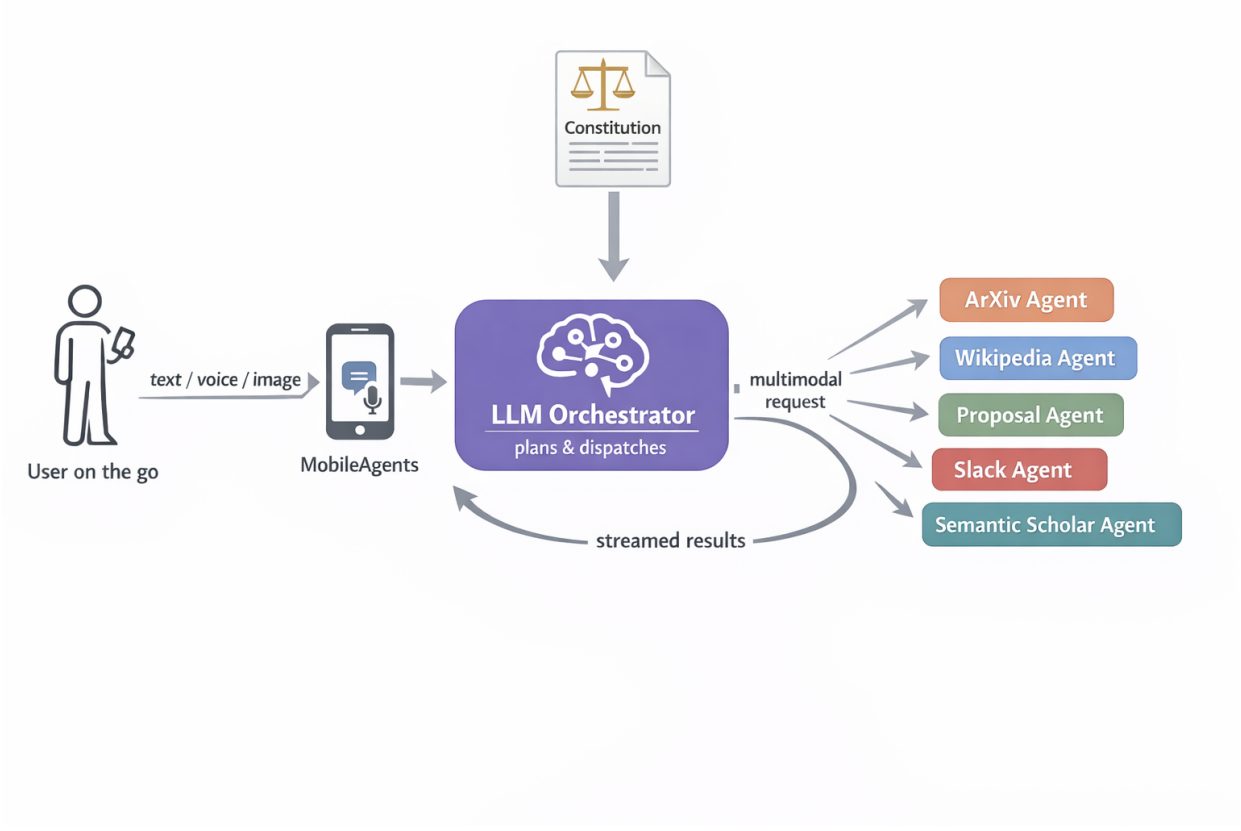


Figure 2: MobileAgents system overview. A user on the go issues a multimodal request (text, voice, or image) from their phone. The LLM Orchestrator decomposes it into a dependency-aware plan and dispatches tasks across the user's personal agent pool (ArXiv, Wikipedia, Proposal, Slack, Semantic Scholar). Results stream back to the mobile client in real time.

2.2 Design Rationale

To begin, I drew on human-centred AI principles from the lectures (reliability, safety, transparency) [21] and looked online for interaction guidelines that emphasise: (a) setting expectations, (b) enabling

efficient correction/repair, and (c) communicating uncertainty [1]. I also used Value Sensitive Design (VSD) [8] as inspiration into how to translate stakeholder values into concrete interaction requirements.

V1: Fast "It's a simple task; it should just do it instead of asking for permission."

V2: Control "There should be guardrails and a control mechanism for sensitive tasks."

V3: Mobility: "I should be able to reach my agents hands-free."

V4:

From these values, I drafted some initial resulting design implications, using methods from VSD [8] and related work [25]:

D1: Design a setting giving the agent full autonomy for lowr**D2:** Propose a plan and ask for confirmation before executing in plan mode; Propose plan and render a real-time execution graph showing active processes (Figure 3).

D3: Support customization of orchestration policy and output stylization (Appendix D).

D4: Provide the ability to turn agents on and off in the interface (Figure 5).

D5: Support hand-free voice-only mode (Figure 1c).

D6: Support images for quick snaps of environmental context (whiteboards, notes, etc) (Figure 1b).

I used these to begin thinking about the prototype, which I called 'Mobile Agents'. We support D1 and D2 via a control setting that includes Blackbox, Plan, and Transparency modes (Figure 3; Table 1). In. Blackbox gives the system full autonomy to execute tasks. Planning mode constructs a structured plan and asks the user for permission before executing. Transparency mode does everything Plan mode does and provides a real-time graph of execution traces for full visibility into the task. Additional control is implemented through D3 and D4. D3 is implemented through a natural-language constitution provito the Orchestrator agent in the system prompt to guide orchestration behavior and output style. D5 and D6 are exercised through the modality setting. We support text and images, native voice input/output (push-to-talk), and multimodal (speech-to-text, text, and images) (Figure 1). The text input is the baseline interaction with the multi-agent system and supports all control settings, mobile, and visualizations. Images are added to the text and are useful for capturing physical context (e.g., whiteboard, notes) using the device's camera. The voice-only setting enables hands-free use but does not support plan and transparency controls. Multimodal is similar to text and images mode but also supports speech input that is converted to text, which does not limit plan and transparency modes.

Table 1: The three transparency configurations compared.

Configuration	What the user sees and can do
C1: Black Box	Query in, final answer out. Agents auto-execute.
C2: Plan Preview	Orchestrator's task plan.
C3: Full Transparency	Plan with a real-time graph, and checkpoint approvals

2.3 Accessibility and Inclusion

HCAI demands inclusivity across age, language, culture, and ability levels [5]. MobileAgents supports English, Chinese, Arabic with full RTL layout, and Bulgarian. It has dark, light, and auto color themes. Icon buttons contain ARIA labels. Users with different abilities and devices can utilize the product through multimodal inputs of text, voice, and images, with neither being a point-of-failure. The design was chosen with personalization in mind and to not impose a single interaction pattern.

3 Methods: Research Question, Evaluation Methods, and Metrics

3.1 Research Question

RQ1: Is there an efficiency/control tradeoff between the Blackbox, Plan, and Transparency modes?

RQ2: Does the support of multiple modalities (text, voice, images) improve the mobile-first experience?

RQ3: Are users able to tune the orchestration behavior through the constitution and on/off switches in the agent management tab (Figure 5)?

3.2 Study Design

The study runs in three sequential phases, each targeting one RQ. All phases use a within-subjects design; order effects are controlled by counterbalancing with a Latin square.

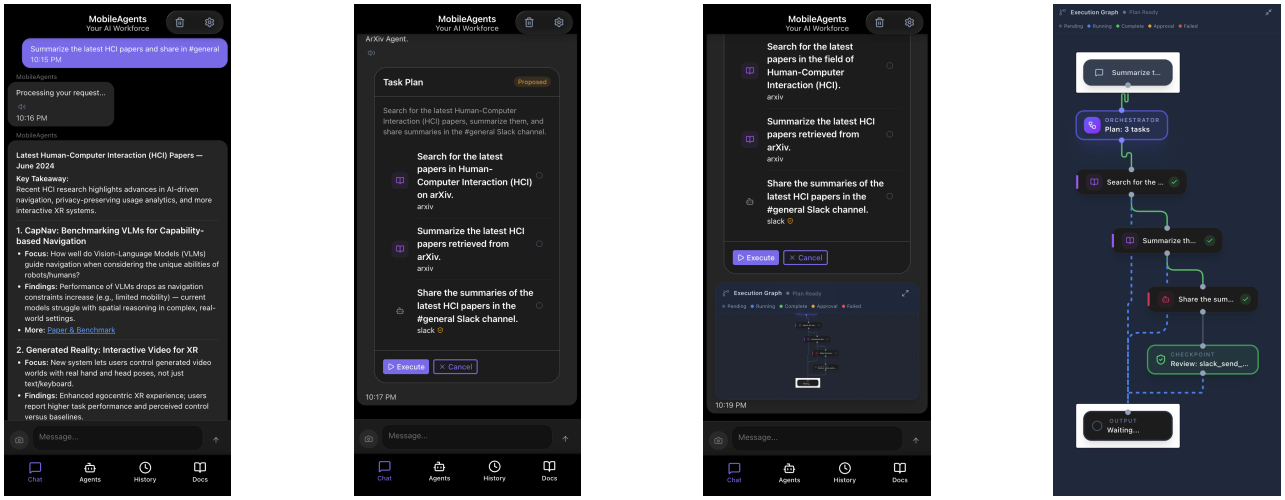
Phase 1 — Transparency modes (RQ1). Each participant completes the same research task (find and compare two recent papers on an assigned topic) under all three transparency configurations—Blackbox (C1), Plan Preview (C2), and Full Transparency (C3)—in counterbalanced order. The task is held constant so that any difference in time, approval behaviour, or subjective ratings is attributable to the configuration, not the task.

Phase 2 — Modality comparison (RQ2). Each participant completes a fixed multi-step workflow (find related work and draft a short proposal) using each of the three input modalities in counterbalanced order: (M1) text only, (M2) image (photograph a printed abstract), (M3) voice dictation. All three runs use Plan Preview (C2) as the transparency baseline so that modality is the only variable. This controls for configuration effects identified in Phase 1.

Phase 3 — Agentic customisation (RQ3). Participants receive an open-ended scenario: "Configure the system so that it never sends Slack messages without showing you a draft first, and remove any agents you would not want running automatically." They are then asked to run a multi-step query and observe whether the system respects their settings. This phase is not counterbalanced; it is a single exploratory session designed to surface whether participants can locate, understand, and apply the constitution editor and agent on/off toggles (Figure 5).

Tasks.

- **T1** (Phase 1 & 2 baseline): Type a text query to find and compare two recent papers on a given topic. Exercises ArXiv and Semantic Scholar agents.
- **T2** (Phase 2, M2): Photograph a printed paper abstract with the device camera and ask the system to find related



(a) C1: Black Box (b) C2: Plan Preview (c) C3: Full Transparency (d) Live DAG (C3 detail)

Figure 3: The three transparency configurations and the C3 graph. (a) Black Box (b) Plan Preview (c) Full Transparency (d) Graph Close-up of Full Transparency

work and draft a proposal. Exercises image analysis, ArXiv, Wikipedia, and Proposal agents.

- **T3** (Phase 2, M3): Use voice input to dictate a request to summarise results and send them to a Slack channel. Exercises voice transcription and the Slack agent with checkpoint approval.
- **T4** (Phase 3): Edit the orchestrator constitution and toggle at least one agent off, then rerun T1 and verify the change in behaviour.

3.3 Participants

We recruited 13 participants from a university graduate programme, spanning age ranges 18–21 ($n = 2$), 22–25 ($n = 2$), 26–30 ($n = 3$), 31–35 ($n = 4$), and 55+ ($n = 2$). All had experience with LLM-based tools; none had used a multi-agent system. Participation was voluntary.

3.4 Evaluation Methods

Traditional usability metrics (task completion, click-through rates) are insufficient for AI systems, where evaluation must also capture trust calibration, mental model accuracy, perceived agency, and error tolerance [4]. We therefore adopt a mixed-methods design combining objective performance measures, validated survey scales, and qualitative user involvement. Table 2 maps each method to the RQ it primarily addresses.

Performance logging (quantitative). The server logs timestamps for each event (input submitted, plan generated, approval clicked, execution complete), enabling precise measurement of task completion time, plan inspection time, and approval/rejection counts per session. Agent toggle and constitution edit events are also logged for Phase 3.

Post-task surveys (quantitative). After each Phase 1 condition, participants complete a 7-point Likert survey covering *trust*, *perceived control*, and *satisfaction* (full items in Appendix M). Trust

Table 2: Evaluation methods by research question.

RQ	Methods	Type
RQ1	Task time, approvals/rejections, trust & control survey, think-aloud	Quant + Qual
RQ2	Task time per modality, NASA-TLX, SEQ, modality preference survey, think-aloud	Quant + Qual
RQ3	Constitution edit count, agent toggle count, task success after customisation, interview	Quant + Qual

items draw on Jian et al.'s [11] Trust in Automation scale. After each Phase 2 modality condition, participants additionally complete the NASA Task Load Index [?] (six subscales: mental demand, physical demand, temporal demand, performance, effort, frustration) to capture cognitive load differences across modalities, and the Single Ease Question (SEQ, 1 item, 7-point) [?] for a quick per-task usability signal. After Phase 3, participants rate the constitution and agent toggle on a 3-item perceived controllability scale.

Think-aloud sessions (qualitative). Participants verbalise their thoughts during all phases. Sessions are screen-recorded and the researcher takes timestamped notes on hesitations, unexpected interactions, plan inspection behaviour, and modality switches. This method surfaces *in-situ* mental model accuracy and trust calibration [4].

Semi-structured interview (qualitative). A 15-minute post-study interview (questions in Appendix N) explores transparency preferences across RQ1 conditions, modality comfort and context sensitivity for RQ2, and whether the constitution and agent controls felt understandable and useful for RQ3. Interview and think-aloud

data are analysed together using Braun and Clarke's [2] reflexive thematic analysis (six phases: familiarisation, coding, theme generation, review, definition, write-up).

3.5 Metrics

RQ1 — Efficiency/control tradeoff:

- Task completion time (s): first input to final response, per configuration.
- Plan approval rate: proportion of plans approved vs. rejected or modified.
- Number of checkpoint interventions per session.
- Trust (Jian et al. [11]), perceived control, and satisfaction (Likert, 1–7).
- Qualitative themes on when participants felt over- or under-informed.

RQ2 — Multimodal mobile experience:

- Task completion time per modality condition.
- NASA-TLX composite score and subscale scores per modality.
- Single Ease Question (SEQ) per task.
- Modality preference ranking and stated switching rationale (interview).
- Qualitative themes on contextual appropriateness of each modality.

RQ3 — Orchestration tuning:

- Constitution edit count and semantic category of edits (constraint, style, routing).
- Agent toggle count and which agents were disabled.
- Task success after customisation: binary measure of whether the system respected the user's constitution rule and agent configuration during T4.
- Perceived controllability scale score (3-item Likert).
- Qualitative themes on discoverability, mental model of the constitution, and perceived ownership.

4 Results

4.1 Quantitative Findings

Table 3 summarises primary survey scores across the three transparency configurations ($N = 13$). We report means and population SD alongside 95% t-based confidence intervals. For within-subjects ordinal data we use the Friedman test; significant Friedman results are followed by pairwise Wilcoxon signed-rank tests with Bonferroni correction ($k = 3$, adjusted $\alpha = .017$) and Cohen's d effect sizes (Table 5).

Table 3: Mean (SD) primary survey scores (1–7 Likert) per transparency configuration.

Metric	C1: Black Box	C2: Plan Preview	C3: Full Transp.
Trust	4.3 (1.7)	5.0 (1.5)	5.8 (1.0)
Control	3.2 (2.1)	5.4 (1.3)	5.7 (1.1)
Satisfaction	4.4 (2.3)	3.9 (1.1)	5.8 (1.1)

Supplementary items are reported in Table 4.

Table 4: Mean (SD) supplementary survey scores (1–7 Likert) per transparency configuration.

Metric	C1: Black Box	C2: Plan Preview	C3: Full Transp.
Expect. Setting	4.2 (1.7)	4.8 (1.1)	6.0 (0.9)
Repairability	3.3 (2.2)	3.8 (1.7)	4.9 (1.2)
Explainability	4.4 (1.7)	5.5 (1.2)	6.0 (1.2)
Privacy Comfort	4.0 (2.2)	5.2 (1.3)	5.5 (1.4)

Trust and control increased monotonically from C1 to C3 (Trust: 4.3 (1.7) → 5.0 (1.5) → 5.8 (1.0); Control: 3.2 (2.1) → 5.4 (1.3) → 5.7 (1.1)). Friedman tests revealed significant within-subjects differences for Control ($\chi^2(2) = 6.43$, $p = .040$), Satisfaction ($\chi^2(2) = 6.53$, $p = .038$), Expectation Setting ($\chi^2(2) = 6.26$, $p = .044$), and Repairability ($\chi^2(2) = 7.96$, $p = .019$); Trust was not significant ($p = .089$). Post-hoc Wilcoxon tests (Table 5) showed control gains from C1 to both C2 and C3 were significant (C1→C2: $p_{\text{bon}} = .041$, $d = -0.88$; C1→C3: $p_{\text{bon}} = .012$, $d = -1.03$). Satisfaction dipped at C2 (3.9 (1.1)) relative to C1 (4.4 (2.3)) before peaking at C3 (5.8 (1.1)), with the C2→C3 jump significant ($p_{\text{bon}} = .012$, $d = -1.16$), suggesting plan approval without the execution graph added friction; the live DAG in C3 resolved this. Expectation setting improved significantly from C1→C3 ($p_{\text{bon}} = .021$, $d = -0.92$) and C2→C3 ($p_{\text{bon}} = .032$, $d = -0.90$). For everyday use, 8/13 preferred C2, 4/13 C1, and 1/13 C3; for high-stakes tasks, 8/13 preferred C3 and 5/13 C2.

Table 6 reports RQ2 metrics across modalities.

Table 5: Post-hoc pairwise Wilcoxon signed-rank tests (Bonferroni-corrected, $k = 3$) and Cohen's d for primary RQ1 metrics. $^*p_{\text{bon}} < .05$.

Metric	Pair	W	p_{bon}	d
Trust	C1-C2	16.0	0.756	-0.33
Trust	C1-C3	9.5	0.120	-0.69
Trust	C2-C3	18.0	0.621	-0.44
Control	C1-C2	6.0	0.041*	-0.88
Control	C1-C3	3.0	0.012*	-1.03
Control	C2-C3	26.5	1.000	-0.16
Satisfaction	C1-C2	29.0	1.000	0.15
Satisfaction	C1-C3	10.0	0.234	-0.58
Satisfaction	C2-C3	2.0	0.012*	-1.16

Voice scored highest on ease (SEQ 5.4 (1.6)), followed by image (4.5 (2.1)) and text (4.2 (2.0)). Despite being rated easiest, voice carried the highest workload (TLX composite 4.6 (0.6)), driven by temporal demand (5.0 (1.6)) and effort (5.6 (1.8)); frustration was actually lowest for voice (3.4 (2.1)). Image was rated most natural for the task (4.9 (1.7)), while voice had the lowest public comfort (2.8 (1.3)). The Performance subscale (reversed: 1=Perfect, 10=Failure) was worst for text (7.3 (1.5)) and best for voice (5.8 (2.1)), reflecting that voice-initiated tasks were handled well by the system. Friedman tests found no significant modality differences in SEQ ($\chi^2(2) = 4.44$, $p = .109$) or TLX composite ($\chi^2(2) = 3.85$, $p = .146$), and no pairwise comparison survived Bonferroni correction (all

Table 6: Mean (SD) SEQ (1–7), NASA-TLX composite and subscales (1–10), and comfort ratings (1–7) per modality.

Metric	M1: Text	M2: Image	M3: Voice
SEQ (ease, 1–7)	4.2 (2.0)	4.5 (2.1)	5.4 (1.6)
TLX composite	4.2 (0.6)	4.1 (0.9)	4.6 (0.6)
Mental	5.5 (2.6)	5.7 (2.9)	5.6 (2.6)
Physical	1.8 (0.9)	1.8 (0.9)	2.0 (0.8)
Temporal	2.6 (1.4)	2.8 (1.7)	5.0 (1.6)
Performance	7.3 (1.5)	6.6 (1.5)	5.8 (2.1)
Effort	3.7 (1.9)	4.2 (2.1)	5.6 (1.8)
Frustration	4.5 (1.6)	3.5 (1.5)	3.4 (2.1)
Felt natural (1–7)	4.2 (1.1)	4.9 (1.7)	4.1 (2.0)
OK in public (1–7)	4.0 (1.6)	3.3 (1.4)	2.8 (1.3)

$p_{\text{bon}} > .37$), likely reflecting limited statistical power at $N = 13$. Preferred modality for everyday mobile use: 6/13 Text, 4/13 Voice, 2/13 Multimodal, 1/13 Image.

Table 7 reports RQ3 items assessing orchestration customisation.

Table 7: Mean (SD) customisation items (1–7 Likert), RQ3.

Item	Mean (SD)
Discoverability	3.8 (1.6)
Mental Model	4.5 (1.8)
Effectiveness	5.2 (1.4)
Controllability	5.7 (1.0)
Ownership	4.8 (1.1)
Adoption	4.2 (1.5)

One-sample Wilcoxon tests against the scale midpoint (4) showed that Controllability ($W = 0$, $p = .001$), Effectiveness ($W = 6$, $p = .016$), and Ownership ($W = 9$, $p = .036$) all rated significantly above neutral; Discoverability ($p = .545$) and Adoption ($p = .717$) did not differ from neutral. Controllability (agent on/off) scored highest (5.7 (1.0)), while constitution Effectiveness was lower (5.2 (1.4)), indicating that users could operate the controls but were less certain that edits changed system behaviour as expected. Discoverability (3.8 (1.6)) was the lowest-rated item, reflecting that the constitution editor remained hard to find without guidance. Overall trust in the system was 5.1 (1.4) and adoption intent was 5.0 (2.0). The most trusted feature was *Live execution graph (seeing agents work in real time)*; the most cited friction point was *Lack of undo functionality*.

4.2 Qualitative Themes

Reflexive thematic analysis of think-aloud and interview data yielded four themes:

Theme 1: Intervention trumps observation. Participants valued the ability to *act*—approving, rejecting, or modifying a plan—more than passively watching execution. One participant noted: "Seeing the graph is cool, but what actually makes me trust it is that I can stop it before it sends the Slack message." This directly validates Amershi et al.'s [1] G2 ("support efficient correction") and G18 ("support efficient dismissal"). The quantitative data reflects this: control scores rose sharply from C1 (3.2) to C2 (5.4) even though C2 adds only a plan card, not a live graph—suggesting that the *ability to intervene* matters more than execution visibility.

Theme 2: Plan Preview as the 'sweet spot.' Satisfaction dipped at C2 (3.9) relative to C1 (4.4) before peaking at C3 (5.8)—a C2→C3 difference that was statistically significant ($p_{\text{bon}} = .012$, $d = -1.16$). This suggests the plan approval step alone created friction; the full execution graph in C3 appears to have justified the overhead. For high-stakes tasks (Slack sends), participants preferred C3 for the checkpoint approval, but defaulted to C2 for routine queries. This maps onto Shneiderman's [21] HCAI framework: users want the *option* of high control but adapt their preferred level to task risk.

Theme 3: Multimodal input essential for mobile, but privacy-gated. Voice had the highest TLX composite (4.6) compared to text (4.2), driven by temporal demand (M=5.0) and effort (M=5.6); frustration was lowest across all modalities for voice (M=3.4). Participants strongly valued voice for hands-free, on-the-go use, but consistently avoided it in shared or public spaces. Image capture was praised for the photographed-abstract task: "I just pointed my phone at the paper and it figured out what to search for." The consent dialog was well received but participants wanted per-session rather than permanent consent.

Theme 4: Constitution discoverability is a barrier; agent toggles are not. Agent on/off controllability scored 5.7/7 while constitution discoverability scored 3.8/7, indicating that toggles are intuitive but the free-text constitution field is harder to find and interpret. The most common constitution edit type was modified tone/style (e.g., "always respond formally"). Participants who did edit the constitution reported a stronger sense of ownership: "It felt like the system was actually mine."

4.3 Sample Interaction Excerpts

Mobile camera + Full Transparency (C3): A participant photographed a printed paper abstract with their phone camera and said "find related work and draft a proposal." GPT-4.1 vision extracted the topic; the orchestrator produced a 3-step plan across three agents: (1) ArXiv search, (2) Wikipedia background, (3) Proposal generation (depending on 1 and 2). The participant inspected the plan card, said "that makes sense," and approved. The live DAG animated steps 1 and 2 in parallel, then step 3 sequentially. The participant commented: "I just pointed my phone at it and it figured out the whole workflow. I can see it's doing both searches at once—that's efficient."

Voice + Plan Preview (C2): A participant used voice input while standing to say "summarise what we found and send it to the team Slack channel." Whisper transcribed the request; the orchestrator planned a 2-step sequence (synthesise → Slack send, with a checkpoint). The plan card appeared; the participant tapped approve. When the checkpoint appeared before the Slack send, they said: "Good—I want to see the message before it goes out." After approving, the message was sent and confirmed. This illustrates the mobile voice workflow: hands-free command, quick plan review, and human-in-the-loop approval for external actions.

5 Discussion

5.1 Ethical, Social, and Value-Related Reflection

Automation vs. autonomy on mobile. Tomašev et al. [23] frame AI delegation as involving transfer of not just tasks but *authority, responsibility, and accountability*—requiring clear role definitions and trust-building mechanisms between human and AI participants. The bring-your-own-agents model amplifies this delegation challenge [21]: users may register agents with powerful capabilities (sending messages, posting to services) that they control from a small screen while distracted. Black Box mode maximises efficiency (V1) but eliminates user agency (V2)—risky when agents act on the user's behalf across external services. This directly contravenes HCAI's transparency and control principle, which holds that users must be able to understand how decisions are made and intervene when needed [5]. Full Transparency restores control but adds time cost that undermines the speed advantage of mobile voice/camera input. Our finding that Plan Preview balances these values suggests that *selective* transparency—showing the plan but not every execution detail—is the pragmatic default for mobile agent control. In VSD terms, this is a *value flow* [7]: a design choice that advances multiple values simultaneously (efficiency *and* control), whereas Black Box creates a *value dam* that blocks control entirely. This connects to HCAI's contextual awareness principle: the right level of transparency depends on the user's environment and task risk, not a universal setting.

Privacy and consent in mobile contexts. Mobile multimodal input intensifies privacy concerns (V3). Users record voice on buses and photograph whiteboards in shared offices—contexts where bystanders may be captured and ambient noise may leak sensitive content. Sending this data to OpenAI raises data governance concerns. Friedman et al.'s [7] VSD method of *informed consent modelling* demands that consent be specific, revocable, and contextually appropriate. Our consent dialog addresses informed consent but does not fully meet this standard: it does not solve data minimisation, since audio is transcribed server-side, meaning raw audio traverses the network. A production system should transcribe on-device or use end-to-end encryption. The agent-agnostic architecture adds a further dimension: if a user registers an agent that connects to a sensitive service (e.g., a CRM or health record), the privacy surface expands beyond what the platform can govern—a concern that VSD's *scalable information dimensions* method [7] is designed to surface, by probing how scale-related factors (data granularity, pervasiveness, number of connected services) impact values differently at different levels of adoption. The tertiary stakeholder (IT) would require per-agent data policies and audit logs that our prototype lacks. Responsible AI principles—anticipation, reflection, inclusion, and accountability [18]—demand that we *anticipate* these risks at design time rather than deferring them to deployment [24].

Trust calibration and over-reliance. HCAI frames trust as built over time through transparency, predictability, reliability, and consistency—a function of perceived benevolence, competence, and integrity [5]. Higher transparency correlated with higher trust, but *trust calibration*—whether trust matches actual system performance—is the real goal. Two participants in C3 accepted agent outputs without checking sources because "I could see the whole process, so it must be right." This *automation bias* [14] is an unintended consequence of transparency: seeing the mechanism can substitute for evaluating the output. Buçinca et al. [4] warn that subjective measures of explainability can be misleading—users may *feel* they understand the system without actually being able to detect errors. Tang et al.'s [22] "Human Tool" framework offers one promising direction: rather than humans always directing AI, the system can proactively *request* human input when it detects uncertainty—structured schemas define what humans can contribute (capabilities, domain knowledge, decision authority), enabling agents to invoke human participation intelligently. Future designs should pair process visibility with such active engagement mechanisms rather than passive observation.

Inclusivity and access. Voice input lowered barriers for one participant with a repetitive strain injury, but the English-only Whisper model disadvantaged a non-native speaker whose accent was poorly transcribed. The four-language UI helps, but speech recognition remains an equity gap. Amershi et al.'s [1] G13 ("mitigate social biases") applies directly: if voice is the primary mobile modality but ASR models perform unevenly across accents, the system systematically disadvantages certain user groups. Costanza-Chock's [6] design justice lens sharpens this: those most harmed by ASR bias—non-native speakers, speakers of under-resourced languages—are precisely the users whose needs should be centred in design, yet they are typically absent from training data and user studies alike. Multimodal systems must offer robust text pathways as a genuine fallback, not just a nominal alternative—van de Poel [25] terms this a *design for value* obligation, where inclusivity must be a design constraint rather than an afterthought.

Societal impact and the value alignment problem. The combination of mobile accessibility and agent-agnostic orchestration lowers the barrier to delegation dramatically: a voice command while walking can trigger a multi-agent workflow that searches, synthesises, and messages on the user's behalf. This frictionlessness risks deskilling—if users habitually delegate without engaging with intermediate results, they may not develop critical synthesis skills. It also risks *action at a distance*: an agent sends a message or posts content while the user is distracted, with consequences they only discover later. The agent-agnostic architecture intensifies the *value alignment problem*: because users register arbitrary agents with arbitrary capabilities, the platform cannot guarantee that agent behaviour aligns with user values at design time. The editable constitution and checkpoint gates are our mitigation, but they shift the burden of alignment onto the user—a tension that responsible AI frameworks flag as a risk of "ethics washing" [18]. Sadek et al.'s [18] scoping review of responsible AI in practice warns that RAI risks becoming a "McDonald's-isation" of ethics: oversaturated with abstract principles but lacking enforceability and situated accountability. Our design attempts to avoid this by making ethical safeguards concrete and actionable (checkpoints, constitution rules) rather than abstract; however, the risk of passive compliance remains if users never edit the constitution or routinely auto-approve checkpoints. Reflective transparency (Plan Preview with mandatory checkpoints for external actions) mitigates this, echoing Yang et al.'s [26] framing of human–AI interaction as a relationship that should support growth, not bypass it.

5.2 Limitations, Future Work, and Conclusion

Limitations. The sample ($N = 10$) is small and drawn from one university, limiting generalisability. The within-subjects design risks carry-over effects despite counterbalancing. Tasks were simulated (participants did not have real research deadlines), which may inflate satisfaction scores and limit ecological validity. The study was cross-sectional; Yang et al. [26] emphasise that human–AI relationships evolve over time, so trust calibration may shift substantially with extended use. Our VSD analysis applied the tripartite framework [7, 8] but did not employ the full breadth of Friedman et al.'s [7] 17 catalogued VSD methods—we did not use envisioning cards, value scenarios, value sketches, scalable information dimensions, or multi-lifespan timelines to explore long-term and indirect value implications. Sadek and Mougenot [19] identify this as a common challenge in VSD for AI: practitioner interviews reveal that time constraints and the opacity of AI behaviour make systematic value elicitation difficult in practice. The tertiary stakeholder group (IT administrators) was not directly consulted, and our participatory engagement [20] was limited to informal conversations rather than structured co-design workshops—risking what participatory design literature warns against as tokenistic consultation rather than genuine collaboration.

Future work. (1) A longitudinal deployment (diary study) where users register their own agents and use MobileAgents as their daily mobile control surface, to study how trust calibration and transparency preferences evolve over time—critical because, as HCAI research emphasises, human–AI relationships are dynamic, not static [5, 26]. (2) Adaptive transparency that automatically adjusts based on task risk (e.g., auto-elevate to Full Transparency when an agent will perform an external action), operationalising HCAI's contextual awareness principle. (3) On-device speech transcription and local image analysis to address the mobile privacy gap. (4) Counterfactual testing [4] where participants see alternative AI plans and evaluate whether their mental models accurately predict system behaviour. (5) Participatory co-design workshops [20] with diverse stakeholder groups—including IT administrators and non-native English speakers—using VSD envisioning cards, value scenarios, and multi-lifespan timelines [7] to surface long-term value implications (e.g., how habitual agent delegation reshapes professional practices across generations). (6) Applying the value-sensitive action-reflection model [7] in iterative design sprints, embedding reflective value prompts into each design cycle to guard against value drift as features accumulate. (7) Integration with messaging-based agent platforms such as Clawdbot [?], which already reach hundreds of thousands of users via WhatsApp and Telegram but lack plan visibility and approval mechanisms. MobileAgents's transparency layer (plan cards, checkpoint gates, live DAG) could be delivered as rich message cards within these messaging interfaces, bringing human-centred oversight to the channels where users already control their agents—without requiring them to switch to a separate app. (8) Adopting emerging interoperability standards to fully realise the bring-your-own-agents vision. The Model Context Protocol (MCP) [?], an open standard introduced by Anthropic and now maintained under the Linux Foundation, defines a client–server protocol for connecting AI systems to external tools and data sources; MobileAgents could act as an MCP client, allowing users to point the orchestrator at *any* MCP-compatible tool server—databases, APIs, local file systems—and have it appear as a registered agent capability, with no platform-specific integration required. Google's Agent2Agent (A2A) protocol [?] complements this at the agent level: agents advertise their capabilities via JSON "Agent Cards," enabling dynamic discovery across networks. Together, MCP (for tools) and A2A (for agents) would let MobileAgents users bring not only their own agents but their own tool ecosystems, connecting to arbitrary services from their phone while MobileAgents's transparency layer ensures human oversight over what these tools do. (9) Cross-device continuity, where a plan started via voice on mobile can be reviewed and refined at a desktop.

Conclusion. MobileAgents demonstrates that personal AI agents can be made controllable from a mobile device through multimodal input (text, voice, camera) and an agent-agnostic orchestration layer. The bring-your-own-agents architecture means users are not locked into a fixed agent set—they plug in whatever tools matter to them and manage them from their phone. Configurable transparency lets users match oversight to context: Plan Preview for quick mobile use, Full Transparency for high-stakes actions. The ability to *intervene*—not merely observe—drove trust more than visual complexity. Consent-gated multimodality respected privacy in mobile contexts, though speech recognition equity and automation bias remain open challenges. These findings contribute to the design space of mobile, multimodal, user-controllable multi-agent AI systems.

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A Architecture Diagram

Figure 4 illustrates the end-to-end data flow across the three system layers.

[Insert architecture diagram here]

Figure 4: MobileAgents system architecture. The mobile React client communicates with the FastAPI server over REST and SSE. The server delegates planning to a CrewAI orchestrator and streams execution updates back to the client in real time.

B Project Structure

```
agentflow/
+-- package.json          # Monorepo scripts
+-- client/
|   \-- src/
|       +-- App.tsx       # Root (auth, theme, RTL)
|       +-- index.css     # Tailwind + animations
|       +-- state/
|       |   \-- store.ts  # Zustand (state + API)
|       +-- types/        # TS interfaces
|       +-- i18n/         # 4 languages (en/zh/ar/bg)
|       \-- components/
|           +-- Chat/     # Chat + markdown + TTS
|           +-- Agents/   # List + detail + toggle
|           +-- Graph/    # ReactFlow DAG + nodes
|           +-- TaskPlan/ # Plan card + approval
|           +-- Camera/   # Image capture + consent
|           +-- Voice/    # Recording + consent
|           +-- Settings/ # Transparency/theme/i18n
|           +-- Execution/# History + step stream
|           \-- Privacy/  # Consent dialog
+-- server/
|   +-- main.py           # FastAPI + CORS + routers
|   +-- config.py         # Pydantic settings
|   +-- auth.py           # Supabase JWT auth
|   +-- routers/
|   |   +-- chat.py       # POST /api/chat
|   |   +-- execute.py    # POST /api/execute (SSE)
|   |   +-- approve.py    # Checkpoint approval
|   |   +-- agents.py     # Agent CRUD
|   |   \-- conversations.py # Conversation CRUD
|   +-- crew/
|   |   +-- orchestrator.py # Planning + graph
|   |   +-- agents.py      # Backstories
|   |   \-- tools.py       # Tool implementations
|   +-- models/
|   |   +-- messages.py   # ChatRequest/Response
|   |   +-- tasks.py      # TaskPlan/TaskStep
|   |   \-- graph.py      # Graph state models
|   \-- services/
|       +-- execution_tracker.py # Dependency exec
|       +-- image_analyzer.py    # GPT-4.1 vision
|       +-- audio_transcriber.py # Whisper
|       +-- agent_store.py       # JSON agent CRUD
|       +-- db.py                # Supabase DB layer
|       \-- supabase_client.py   # Supabase client
```

C System Prompts

This appendix documents every LLM prompt used in the system. All prompts are reproduced verbatim from the source code.

C.1 Orchestrator Backstory

Injected as the CrewAI agent's `backstory` parameter. The agent's role is "Task Orchestrator" and its goal is "Analyze user requests and create structured execution plans that delegate work to specialized research agents." At runtime, the list of enabled agents and the user-editable constitution are appended dynamically.

```
You are a concise research librarian orchestrating
an academic team. Keep all outputs brief — this is
a mobile app with limited screen space.
```

```
Break requests into the fewest steps needed.
Prefer parallel execution.
Use uncertainty-aware language when appropriate.

Produce structured JSON plans. Do not add
unnecessary steps.

Available agents:
- arxiv: ArXiv Research Analyst —
    capabilities: arxiv_search, arxiv_summarize
- proposal: Research Proposal Strategist —
    capabilities: generate_proposal,
    outline_methodology
- wikipedia: Background Research Specialist —
    capabilities: wiki_search, wiki_summarize
- slack: Slack Messenger —
    capabilities: slack_send_message
    [requires_approval]
- semantic_scholar: Citation Analyst —
    capabilities: semantic_scholar_search,
    semantic_scholar_cite

Constitution (user-defined guidelines):
<user-editable free text appended here>
```

C.2 Planning Task Prompt

Sent to the orchestrator for every user request. Variables in braces are interpolated at runtime. The optional image analysis and audio transcript blocks are only included when the corresponding modality is present. Conversation history is included to resolve ambiguous references (e.g., "do it", "yes", "go ahead").

```
Analyze the following user request and create a
structured execution plan.

User request: {user_message}

[Image analysis: {image_analysis}]
[Audio transcript: {audio_transcript}]
[Conversation history: {history}]

Input modality: {input_modality}
{modality_note}

Available agents:
{agent_descriptions}

Rules:
- Match the user's request to the most appropriate
  agent(s). Agents are NOT limited to research —
  some send messages, perform actions, etc.
- For agents marked [requires_approval], set
  requires_approval=true on that step. For all
  other agents, set requires_approval=false.
- Only return an empty steps list if the request
  truly cannot be handled by ANY available agent
  (e.g. "hello", pure casual chat).
- If steps are independent, leave depends_on empty
  so they can run in parallel.
- If a step needs output from another step, add
  that step's ID to depends_on.
- Always prefer parallel execution when possible.
- Use step IDs like "step_1", "step_2", etc.
- agent_id must be one of: {valid_agent_ids}
- Use exact param names from tool signatures.
  For slack_send_message use params:
  {"channel": "<channel>", "text": "<message>"}
- If image analysis is provided, use that content
  to inform search queries and proposal topics.
- If audio was transcribed, treat the transcript
  as the primary user intent.
- If conversation history is provided, use it to
  resolve ambiguous references.
```

C.3 Synthesis Prompt

Called after all agent steps complete. The system prompt constrains output for mobile readability. The user message aggregates the original request, plan summary, and all step results. Model: GPT-4.1, max tokens: 800.

System prompt:

```
You are a research orchestrator synthesizing agent
results for a MOBILE interface. Responses are read
on small screens.

Guidelines:
- Be concise: aim for 150-300 words max
- Lead with the key finding or answer in 1-2
  sentences
- Use short bullet points, not long paragraphs
- Use markdown: **bold** for emphasis, short
  headings, compact lists
- Keep relevant URLs but don't list more than 3-5
- Do NOT mention the agents or internal execution
  process
- Use uncertainty-aware language: 'based on
  available results' rather than absolute claims
- End with a one-line disclaimer: '*Results may
  not be exhaustive — verify for critical
  decisions.*'
```

User message template:

```
**User request:** {user_message}

**Plan summary:** {plan_summary}

**Agent results:**

### Agent step: {step_description}
{step_result}

### Agent step: {step_description}
{step_result}
...
```

C.4 Image Analysis Prompt

Sent to GPT-4.1 vision when the user provides an image. The image is passed as a base64-encoded JPEG in the `image_url` content block. Max tokens: 300.

```
Analyze this image in the context of a research
assistant. Describe what you see and what research
tasks might be relevant (e.g., paper search,
literature review, proposal generation).
```

`Be concise.`

C.5 Audio Transcription

Audio is recorded via the browser's MediaRecorder API as WebM, base64-encoded, and sent to the server. The server decodes the base64 payload, writes it to a temporary `.webm` file, and passes it to the OpenAI Whisper API (`whisper-1` model). The resulting plain-text transcript replaces the user's text message in the planning prompt.

C.6 Modality Notes

A modality-specific note is injected into the planning context based on the input type:

Table 8: Modality notes injected into the planning prompt.

Modality	Injected note
`text`	"The user typed this request."
`voice`	"The user spoke this request via voice input."
`image`	"The user provided an image along with their request. The image has been analyzed and the analysis is included above."

D Agent Backstories

Each worker agent has a backstory injected as its CrewAI `backstory` parameter. These define the agent's persona and constrain its behaviour during tool execution.

ArXiv Agent: "You are an expert research analyst who monitors arXiv daily. You can search for papers by topic, summarize their key findings, and identify trends across recent publications. You excel at distilling complex papers into clear, actionable summaries."

Proposal Agent: "You are a seasoned research strategist who has helped write dozens of successful grant proposals and research plans. You understand how to identify research gaps, formulate compelling research questions, and outline rigorous methodologies. You synthesize information from literature into coherent proposals."

Wikipedia Agent: "You are a thorough background researcher who uses Wikipedia to gather foundational knowledge on topics. You find relevant articles, extract key concepts and definitions, and provide well-structured overviews that help contextualize research work."

Slack Agent: "You are a communication specialist who sends messages to Slack channels on behalf of the team. You craft clear, well-formatted messages and deliver them to the correct channels. You always confirm which channel to post to and summarize what was sent."

Semantic Scholar Agent: "You are a citation analyst who uses Semantic Scholar to find papers and assess their impact. You search across all academic literature (not just arXiv), retrieve citation counts and influential citation metrics, and help identify the most impactful work in any research area."

E Tool Implementations

Table 9 lists all tool functions available to agents. Each tool is implemented as a standalone Python function and registered in a per-agent tool map. The orchestrator assigns agents to steps; the execution tracker calls the tool function directly (bypassing CrewAI at execution time) for predictable, streamable results.

E.1 Dependency-Aware Parameter Resolution

When a step depends on a previous step's output, the execution tracker performs automatic parameter resolution:

- **`arxiv_summarize`:** If no explicit `paper_url` is provided, arXiv URLs are extracted from the parent step's result using the regex `http://arxiv.org/abs/[\w.]+`.
- **`wiki_summarize`:** Wikipedia titles are extracted from the parent step's Markdown-bold text (`**Title**`).
- **`generate_proposal` / `outline_methodology`:** The step description is used as the topic and all prior results are concatenated as context.
- **`slack_send_message`:** If the `text` parameter contains a placeholder, the previous step's result is substituted. The channel defaults to `#general`.

Parameter aliases are normalised automatically (e.g., `message` → `text`, `recipient` → `channel`, `doi` → `paper_id`).

F Execution Engine

F.1 Two-Phase Architecture

Phase 1 — Planning: The orchestrator (a CrewAI agent backed by GPT-4.1) receives the user's multimodal input and produces a structured `TaskPlan` with a dependency graph encoded in each step's `depends_on` field. A corresponding `ExecutionGraphState` is generated with typed nodes (input, orchestrator, agent, checkpoint, output) and edges. Node positions are computed by the client using the Dagre layout algorithm.

Phase 2 — Execution: The execution tracker iterates the plan in topological order. For each step whose dependencies are satisfied, it: (1) activates incoming edges, (2) sets the node to `running`, (3) calls the tool function directly with resolved parameters, (4) records execution duration, and (5) marks the node as `completed` or `failed`. Steps without mutual dependencies execute in the order they become ready. After all steps complete, the tracker calls the synthesis LLM and emits the final result.

The live DAG for a three-step task is shown as Figure 3(d) in the main body: ArXiv search → summarise → Slack send, with a checkpoint node gating the Slack step.

F.2 SSE Event Protocol

Execution updates are streamed from `POST /api/execute` as Server-Sent Events. Table 10 documents the complete event protocol.

F.3 Checkpoint Approval Flow

When a step has `requires_approval = true`, the execution tracker inserts a checkpoint node between the orchestrator and the agent node. Upon reaching the checkpoint:

(1) The checkpoint node is set to `awaiting_approval`.
(2) A `checkpoint_reached` SSE event is emitted.
(3) The client displays an approval dialog (approve/reject).
(4) The user's decision is sent via `POST /api/approve/{step_id}`.

Table 9: Complete tool registry. Each function is mapped to one or more agents and called directly during execution.

Function	Agent	Parameters	Behaviour
`arxiv_search`	ArXiv	`query`: str, `max_results`: int (default 5)	Queries the arXiv API sorted by submission date (descending). Returns titles, authors, URLs, and truncated abstracts (300 chars). Timeout: 15 s.
`arxiv_summarize`	ArXiv	`paper_url`: str	Fetches a single paper by arXiv ID. Returns title, authors, categories, and full abstract.
`generate_proposal`	Proposal	`topic`: str, `context`: str	Generates a Markdown research proposal with sections: Introduction, Research Questions, Methodology, Expected Contributions, Timeline. Context from prior steps is incorporated.
`outline_methodology`	Proposal	`approach`: str, `domain`: str	Generates a methodology section with Data Collection, Approach Details, Evaluation Metrics, and Reproducibility.
`wiki_search`	Wikipedia	`query`: str	Searches the Wikipedia API; returns up to 5 results with snippets. User-Agent: `MobileAgents/1.0`. Timeout: 10 s.
`wiki_summarize`	Wikipedia	`title`: str	Fetches the Wikipedia REST API summary for a page. Returns title, URL, and extract.
`slack_send_message`	Slack	`channel`: str, `text`: str	POSTs to the Slack API with a Bearer token (`SLACK_BOT_TOKEN`). Returns confirmation or error. Requires checkpoint approval. Timeout: 10 s.
`semantic_scholar_search`	Sem. Scholar	`query`: str, `max_results`: int (default 5)	Queries the Semantic Scholar API. Returns title, year, citation count, influential citation count, authors, URL, and abstract. Timeout: 15 s.
`semantic_scholar_cite`	Sem. Scholar	`paper_id`: str	Fetches citation details for a paper by Semantic Scholar ID, DOI, or `ARXIV:<id>`. Returns citation count, influential citation count, reference count, authors. Handles 404 gracefully. Timeout: 15 s.

Table 10: SSE event types streamed during execution.

Event type	Payload and semantics
`graph_init`	Full `ExecutionGraphState`. Sent once at the start.
`node_status`	`nodeId`, `status` (`running` | `completed` | `failed` | `awaiting_approval` | `approved`), optional `result` and `duration` (ms).
`edge_status`	`edgeId`, `status` (`active` | `completed`).
`checkpoint_reached`	`nodeId`, `stepId`. Emitted when an approval-gated step is ready.
`execution_complete`	Updated `ExecutionGraphState` and synthesised `summary`.
`execution_failed`	`error` message string.

(5) If approved, the checkpoint is marked approved and execution continues. If rejected, the step is skipped.

G Data Schemas

All data models are defined as Pydantic `BaseModel` classes on the server and TypeScript interfaces on the client.

G.1 Request and Response Models

ChatRequest (client → server):

```
{
  "message": "string",
  "image_base64": "string | null",
  "audio_base64": "string | null",
  "input_modality": "text | voice | image",
  "conversation_history": [
    {"role": "user|assistant", "content": "..."}
  ],
  "conversation_id": "string | null"
}
```

ChatResponse (server → client):

```
{
  "message": "string",
  "plan": TaskPlan | null,
  "graph": ExecutionGraphState | null,
  "image_base64": "string | null"
}
```

G.2 Task Plan Models

TaskPlan:

```
{
  "id": "string",
  "summary": "string",
  "steps": [TaskStep],
  "status": "proposed | executing | completed",
  "created_at": "ISO 8601"
}
```

TaskStep:

```
{
  "id": "step_1",
  "agent_id": "arxiv | proposal | wikipedia
              | slack | semantic_scholar",
  "action": "tool function name",
  "description": "human-readable description",
  "params": { "key": "value" },
  "status": "pending | running | completed
            | failed | awaiting_approval",
  "result": "string | null",
  "requires_approval": false,
  "depends_on": ["step_id"]
}
```

G.3 Execution Graph Models

ExecutionGraphState:

```
{
  "task_id": "string",
  "nodes": [GraphNode],
  "edges": [GraphEdge],
  "current_node_id": "string | null",
  "status": "planning | executing
            | completed | failed"
}
```

GraphNode:

```
{
  "id": "string",
  "type": "input | orchestrator | agent
          | checkpoint | output",
  "data": {
    "label": "string",
    "type": "...",
    "status": "pending | running | completed
              | failed | awaiting_approval
              | approved | skipped",
    "agent_id": "string | null",
    "agent_color": "#hex | null",
    "agent_icon": "LucideIconName | null",
    "result": "string | null",
    "duration": "milliseconds | null",
    "input_modality": "text|voice|image | null",
    "timestamp": "ISO 8601 | null"
  },
  "position": { "x": 0, "y": 0 }
}
```

GraphEdge:

```
{
  "id": "e-source-target",
  "source": "node_id",
  "target": "node_id",
  "data": {
    "status": "pending | active | completed",
    "data_preview": "string | null"
  }
}
```

H Agent Registry Schema

Each agent in the persistent registry (`agents_config.json`) follows this schema:

```
{
  "id": "unique_string",
  "name": "Display Name",
  "icon": "LucideIconName",
  "description": "Short description",
  "role": "CrewAI role string",
  "goal": "CrewAI goal string",
  "backstory": "Full system prompt text",
  "capabilities": ["tool_1", "tool_2"],
  "enabled": true,
  "requiresApproval": false,
  "color": "#hex",
  "isOrchestrator": false,
  "constitution": ""
}
```

The `isOrchestrator` flag (only `true` for the orchestrator) prevents it from being disabled or deleted via the UI or API. The `constitution` field is only used by the orchestrator and is appended to its backstory at planning time. The `requiresApproval` flag causes a checkpoint node to be inserted in the DAG before the agent's execution step. The `enabled` flag controls whether the agent appears in the orchestrator's available agent list.

I Client State Management

The client uses a single Zustand store with the following state slices:

Table 11: Zustand store state slices.

Slice	Contents
Navigation	`activeTab`: chat | agents | history | guide
Chat	`messages`: ChatMessage[], `isLoading`: boolean
Planning	`currentPlan`: TaskPlan | null, `graphState`: ExecutionGraphState | null, `isExecuting`: boolean
Agents	`agents`: Agent[], `toggleAgent(id)`, `updateConstitution(id, text)`
Settings	`transparencyLevel`: black_box | plan_preview | full_transparency; `modalityMode`: multimodal | text_image | voice_only; `themeMode`: dark | light | auto; `language`: en | ar | zh | bg
History	`conversations`: Conversation[], `currentConversationId`, `loadConversations()`, `startNewConversation()`, `deleteConversation(id)`

Key methods:

- **`sendChat(message, imageBase64?, modality?, audioBase64?)`**: Posts to `/api/chat` and updates the message list and plan/graph state.
- **`executePlan()`**: Streams SSE events from `/api/execute` and delegates each event to `processSSEEvent()`.
- **`processSSEEvent(event)`**: Applies real-time graph mutations (node status, edge status, checkpoint, completion).

All state is persisted to `localStorage` via a Zustand subscriber on every state change. On page load, the store re-hydrates from the persisted snapshot.

J Internationalisation Keys

The UI is fully localised into four languages: English (en), Chinese Simplified (zh), Arabic (ar), and Bulgarian (bg). The translation system uses a `useT()` hook that returns a function mapping translation keys to localised strings with variable interpolation (e.g., `t('plan.steps', {count: '3'})`). Over 160 keys cover:

- UI chrome: tab labels, settings headers, button labels
- Chat: suggestions, loading states, empty states, markdown rendering
- Agent interface: agent list headers, detail sheet labels, constitution editor
- Task plan: step statuses, approval controls, plan summary
- Execution graph: node labels, legend items, status descriptions
- Privacy: consent dialog text, data handling notices
- Voice: listening, speaking, and thinking state labels

- Authentication: sign in, sign up, error messages

Arabic includes full RTL layout support. The language switcher in settings triggers a re-render of all localised strings and updates the document direction attribute.

K CSS Animations and Theming

K.1 Theme Variables

The interface supports Dark, Light, and Auto (system preference) themes via CSS custom properties:

Table 12: Theme colour tokens (selected).

Token	Light	Dark
`-surface-0`	`#ffffff`	`#000000`
`-surface-1`	`#f2f2f7`	`#1c1c1d`
`-surface-2`	`#e5e5ea`	`#121212`
`-text-primary`	`#000000`	`#ffffff`
`-text-secondary`	`#3c3c43`	`#ffffffcc`

K.2 Graph Animations

Three CSS keyframe animations drive the execution graph visualisation:

- **`pulse-ring`** (1.5 s, cubic-bezier) — Pulsing ring on nodes in the `running` state. Scales from 1× to 1.5× with fading opacity.
- **`dash-flow`** (0.6 s, linear, infinite) — Animated dashed stroke on `active` edges, creating a flow effect along the edge path.
- **`checkpoint-glow`** (2 s, ease-in-out, infinite) — Glowing border on checkpoint nodes in the `awaiting_approval` state.

K.3 Accessibility Styles

All interactive elements (buttons, inputs, textareas, links) receive a `focus-visible` outline: 2 px solid `#7c6aef` with a 2 px offset, ensuring keyboard navigability without affecting mouse/touch interactions.

L Authentication and Data Persistence

L.1 Authentication

The server uses Supabase for authentication. All API endpoints are protected by a `get_current_user` FastAPI dependency that:

(1) Extracts the Bearer token from the `Authorization` header.
(2) Validates it via `supabase.auth.get_user(token)`, which works with both HS256 and ES256 JWT keys.
(3) Returns `{sub: user_id, email: user_email}` or raises HTTP 401.

The client stores the Supabase session token and includes it in all API requests.

L.2 Server-Side Persistence

Conversations, messages, execution records, and approval decisions are persisted to Supabase (PostgreSQL). The database layer (`services/db.py`) provides:

- **Conversations:** Create, list (by user), get, delete. Each conversation stores a title and transparency level.
- **Messages:** Insert and list (by conversation). Each message stores role, content, optional image URL, voice transcript, agent ID, input modality, task plan, and execution graph.
- **Executions:** Create (with plan and graph snapshots) and complete (with status, summary, and step results).
- **Approvals:** Create, resolve (approve/reject with optional comment), and query by step ID.

Agent configurations are stored separately in `agents_config.json` on the server filesystem.

L.3 Client-Side Persistence

As a fallback and for offline access, the client persists the full Zustand store to `localStorage` on every state change. On page load, the store re-hydrates from the persisted snapshot before fetching fresh data from the server.

M Survey Instrument

The full survey is administered in four parts corresponding to the three study phases. All Likert items use a 7-point scale (1 = Strongly Disagree, 7 = Strongly Agree) unless otherwise noted.

Part 0: Background (before study begins)

(1) Age range. *(18–21 / 22–25 / 26–30 / 31–40 / 40+)*
(2) Gender. *(Male / Female / Non-binary / Prefer not to say)*
(3) How often do you use AI assistants (e.g., ChatGPT, Copilot)? *(Daily / Several times a week / Weekly / Monthly / Rarely or Never)*
(4) Have you used a multi-agent AI system before? *(Yes / No / I'm not sure)*
(5) Which AI tools do you use regularly? *(Select all: ChatGPT, Copilot, Gemini, Claude, Cursor, Perplexity, Other)*

Part 1: After each transparency condition (C1, C2, C3) — RQ1

Completed three times, once per configuration. Items 1–3 are primary metrics; items 4–7 are supplementary.

(1) I trusted the system's output enough to act on it. *(Trust)*
(2) I felt in control of what the system did on my behalf. *(Perceived Control)*
(3) I am satisfied with this interaction. *(Satisfaction)*
(4) The system made it clear what it was going to do before doing it. *(Expectation Setting)*
(5) It was easy to correct or stop the system when needed. *(Repairability)*
(6) I understood why the system made the decisions it did. *(Explainability)*
(7) I felt comfortable with the information I shared during this task. *(Privacy Comfort)*
– Any comments about this mode? *(open text, optional)*

After all three conditions:

8. Which transparency mode would you prefer for everyday use? *(C1 Blackbox / C2 Plan Preview / C3 Full Transparency)*

9. Which mode would you prefer for high-stakes tasks (e.g., sending a Slack message on your behalf)? *(C1 / C2 / C3)*
10. Why did you prefer that mode? *(open text, optional)*

Part 2: After each modality condition (M1, M2, M3) — RQ2

Completed three times, once per modality (text, image, voice). The NASA-TLX uses a 1–10 scale (Very Low–Very High); Performance is reversed (1 = Perfect, 10 = Failure).

Single Ease Question (SEQ):

(1) Overall, how would you rate the difficulty of this task? *(1 = Very Difficult, 7 = Very Easy)*

NASA Task Load Index [?]:

2. Mental Demand — How much mental and perceptual activity was required? *(1–10)*
3. Physical Demand — How much physical activity was required? *(1–10)*
4. Temporal Demand — How much time pressure did you feel? *(1–10)*
5. Performance — How successful were you in accomplishing the task? *(1 = Perfect, 10 = Failure)*
6. Effort — How hard did you have to work to accomplish the task? *(1–10)*
7. Frustration — How irritated, stressed, or annoyed did you feel? *(1–10)*

Modality comfort (7-point Likert):

8. Using this modality felt natural for this type of task. *(Modality Comfort)*
9. I would use this modality in a public or shared space. *(Context Appropriateness)*
   - In what situations would you choose or avoid this modality? *(open text, optional)*

After all three modality conditions:

10. Which input modality would you use most often on a mobile device? *(Text / Voice / Image / It depends)*
11. Which modalities felt well-suited to on-the-go use? *(Select all: Text / Voice / Image)*
    - Was there a modality that surprised you – positively or negatively? *(open text, optional)*

Part 3: After the customisation task (Phase 3) — RQ3

(1) I was able to find the constitution editor without help. *(Discoverability)*
(2) I understood what the constitution field does. *(Mental Model)*
(3) Editing the constitution changed the system's behaviour as I expected. *(Effectiveness)*
(4) The agent on/off toggles were easy to understand and use. *(Controllability)*
(5) After customising the system, I felt more in control. *(Perceived Ownership)*
(6) I would use the constitution and agent controls regularly. *(Adoption Intent)*

7. Which type of change did you make to the constitution? *(Constraint / Style / Routing / Did not edit / Other)*
   - Describe what you changed and whether the system behaved as expected. *(open text)*

Part 4: Overall post-study feedback

(1) Overall, I would trust this system to act on my behalf on a mobile device. *(Overall Trust)*
(2) I felt this system respected my privacy. *(Privacy)*
(3) I would use this application in my day-to-day work. *(Adoption Intent)*

4. Which single feature contributed most to your trust? *(Plan Preview / Checkpoint approval / Live DAG / Agent on/off / Constitution / Multimodal input)*
5. Which single feature caused the most friction or concern? *(Plan approval steps / Live graph overwhelming / Voice risky in shared spaces / Camera awkward / Agent controls hard to find / Constitution unclear)*
   - What is the single most important improvement you would make? *(open text)*
   - Any other comments or suggestions? *(open text)*

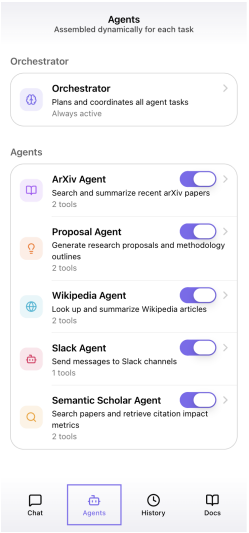
N Semi-Structured Interview Questions

After completing all three configurations, participants were asked:

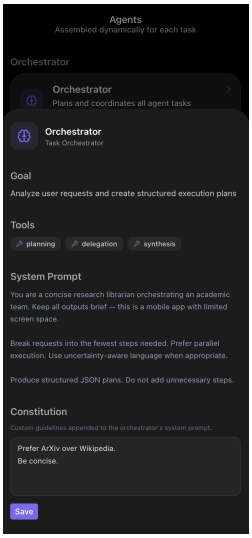
(1) You used three different versions of the system. Can you describe how each felt different?
(2) Which version would you prefer for everyday use? Why?
(3) Was there a moment where you particularly trusted—or distrusted—the system? What caused that?
(4) The system showed you a plan before executing. Did that change how you interacted with it? How?
(5) For the version with the live graph, was seeing the execution progress helpful or distracting?
(6) The system required approval before sending a Slack message. How did that approval step feel? Too cautious? About right?
(7) How did you decide whether to use text, voice, or camera input? Were there situations where you avoided a modality? Why?
(8) Did any of the system's responses feel overconfident or uncertain? How did that affect your trust?
(9) If this system stored your conversation history across sessions, how would you feel about that? What controls would you want?
(10) Can you think of a situation where a system like this could cause harm—to you or someone else?
(11) What is the single most important change you would make to improve this system?

O Interface Screenshots

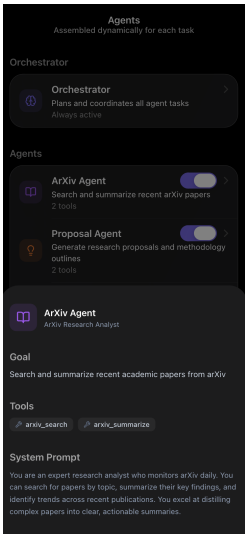
The following screenshots document the complete MobileAgents UI. Figures 5–7 supplement the main-body figures (Figures 1 and 3).



(a) Agents tab (light theme)

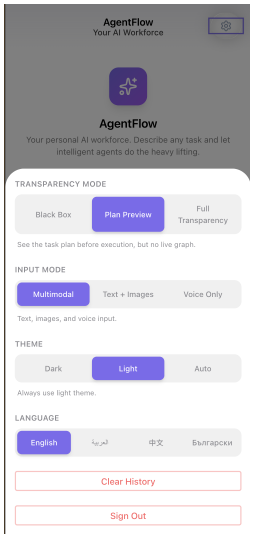


(b) Orchestrator detail with constitution editor

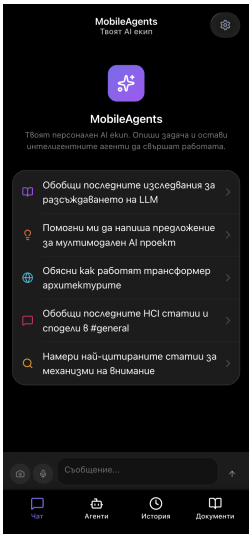


(c) Agent card (ArXiv agent, dark theme)

Figure 5: Agent management screens. (a) The Agents tab lists the orchestrator (always active) and all registered worker agents with per-agent enable/disable toggles. (b) Tapping the orchestrator opens its detail sheet, showing goal, tools, system prompt, and the editable constitution field. (c) A worker agent card shows its goal, tool names, and full system prompt backstory.

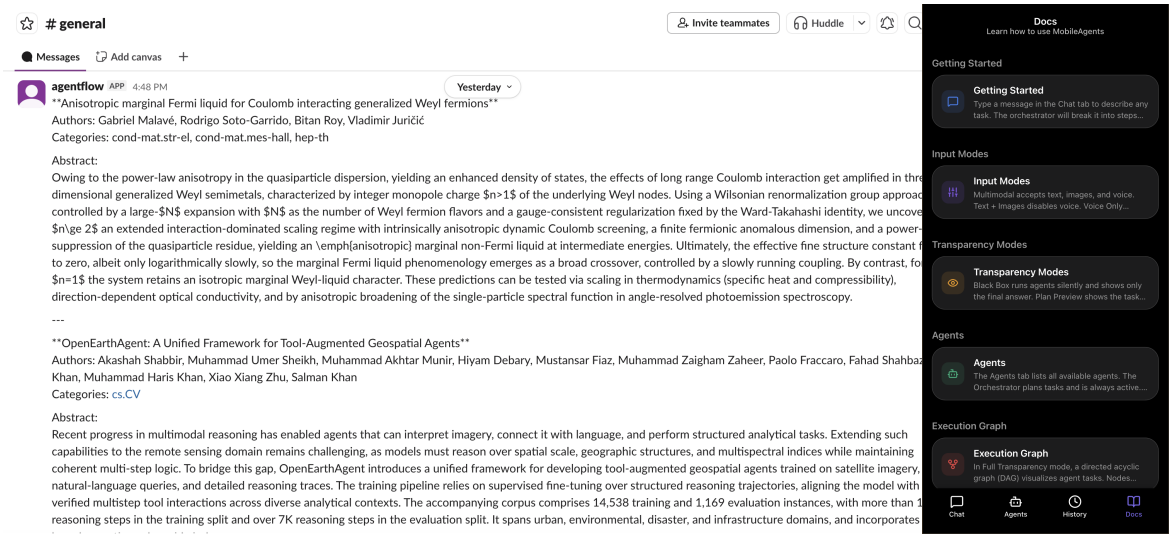


(a) Settings panel



(b) Bulgarian localisation

Figure 6: Settings and localisation screens. (a) Settings exposes all three transparency modes, input modes (Multimodal / Text+Images / Voice Only), theme (Dark/Light/Auto), and language (EN/AR/ZH/BG). (b) Bulgarian localisation: the full UI—including navigation labels and suggested queries—renders in Bulgarian with full RTL support for Arabic (not shown).



(a) Slack output (#general channel) (b) In-app documentation tab

Figure 7: Slack integration and in-app documentation. (a) Result of a Slack send step after checkpoint approval: two arXiv paper summaries posted to #general by the agentflow bot. (b) The Docs tab provides an in-app guide covering input modes, transparency modes, the agents tab, and execution graph; it is available without leaving the app.